

ECE 2036 Lab 4: Farkle Assistant

Due: October 19, 2020 by 11:59PM

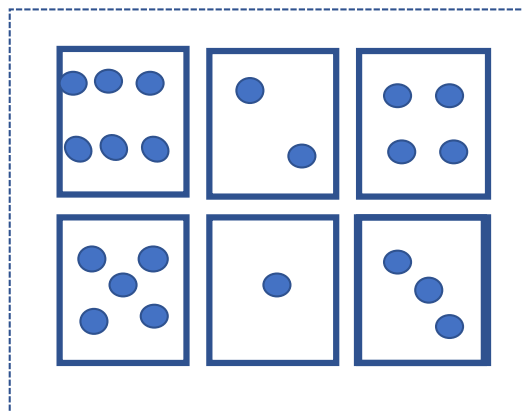
A friend of yours from UGA comes to you about an exciting dice game he has been playing. He is surprised that you have not been playing it, but you sigh and remind him that you are going to school at Georgia Tech...and well you don't have a lot of time to play with dice!

Anyway, he wants to start a company with a "high-tech" version of the game. Surprisingly, he has done the market research, talked to over 200 potential users, and has found a user group who would be interested in this game. He wants to get this "rolling" ASAP to get the game to his early adopters for the holidays. He wants YOU to head the design team for the product for a significant share of equity in the company and profits.

You agree and tell him that you will make a rough prototype of the game over the next week or so to help with the first design of the game. You remember your mbed kit that you bought in ECE2036 and you get cracking!

The dice game is called Farkle and you can see the instructions in Appendix F.

Lab Description: You will use an accelerometer to help simulate the rolling of the dice. In this prototype use will shake your protoboard to roll the dice. The dice will appear on your screen and settle into 6 possible positions. I will give you freedom to choose how you want your dice to appear, but here is a rough sketch of what it could look like.



Remember I want you to design this in an object oriented (OO) way. If you do not, you will lose points (see appendix D)! You need to have perhaps a Die class, a FarkleGame class, etc.. I will talk about various strategies in class using some UML diagrams.

I have emphasized that you should have an incremental design and debugging strategy in designing a system. I encourage you to do this for this assignment. You can create different subtasks, but I have outlined some potential milestones that you must create and verify.

Stage 1: Display your dice. (+20 points)

Please make a short YouTube video documenting this phase to receive these 20 points, and upload the link to Canvas with a label of Stage1.

You have some freedom in what you want your dice to look like. At this stage I would like for you to have objects that you can use that display dice on the screen. You can see some examples in Appendix A- C.

You might think about making a Dice object with various member functions which could include, `rollDice()`, `displayDice(position)`, etc... You might want to have an array of 6 dice in a `FarkleGame` class.

Stage 2: Roll your dice (all six) (+20 points)

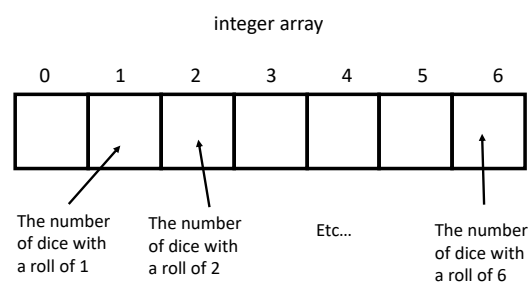
Please make a short YouTube video documenting this phase to receive these 20 points, and upload the link to Canvas with a label of Stage2.

In this stage you will engage your accelerometer to indicate that you would like to roll the dice. You can then use the `rand()` function in a Dice member function called `rollDice()` to roll a die. Be sure you seed the random number generator using `srand(time(0))` once at the beginning of your `main()` function. Do this so that it will generate a new set of random numbers every time you play. Make sure you add a sound to play when you roll the dice! I will let you choose something appropriate.

Stage 3: Calculate the Score of the Dice (+20 points)

Please make a short YouTube video documenting this phase to receive these 20 points, and upload the link to Canvas with a label of Stage3.

Look at the Farkle rules in Appendix F. You will need to calculate the score for the dice roll that you have. I would suggest making an `int` array that represents the histogram of dice values in a roll and using this data to calculate the scores. This `calcRoll()` could be a member function in your `FarkleGame` class (which might also have an array of Dice objects as its data members)



Stage 4: Create a hardware/software system that implements the specifications outlined in the activity (i.e. flowchart) chart in Appendix A. (+20 points added to your total)

Please make a YouTube video documenting this phase to receive a maximum final grade of 80 points, and upload the link to Canvas with a label of Stage4. Please note that if you complete Stage 4 then you do not have to show a video for Stage 1 thru Stage 3. Make sure you show yourself playing your game for at least 20 turns in a variety of different results.

Stage 5: Create a hardware/software system that implements the specifications outlined in the activity chart (i.e. flowchart) in Appendix B. (+10 points added to your total)

Please make a YouTube video documenting this phase to receive a maximum final grade of 90 points, and upload the link to Canvas with a label of Stage5. Please note that if you complete Stage 5 then you do not have to show a video for Stage 1 thru Stage 4. Make sure you show yourself playing your game for at least 20 turns in a variety of different results. Make sure that we see a Farkle condition if possible.

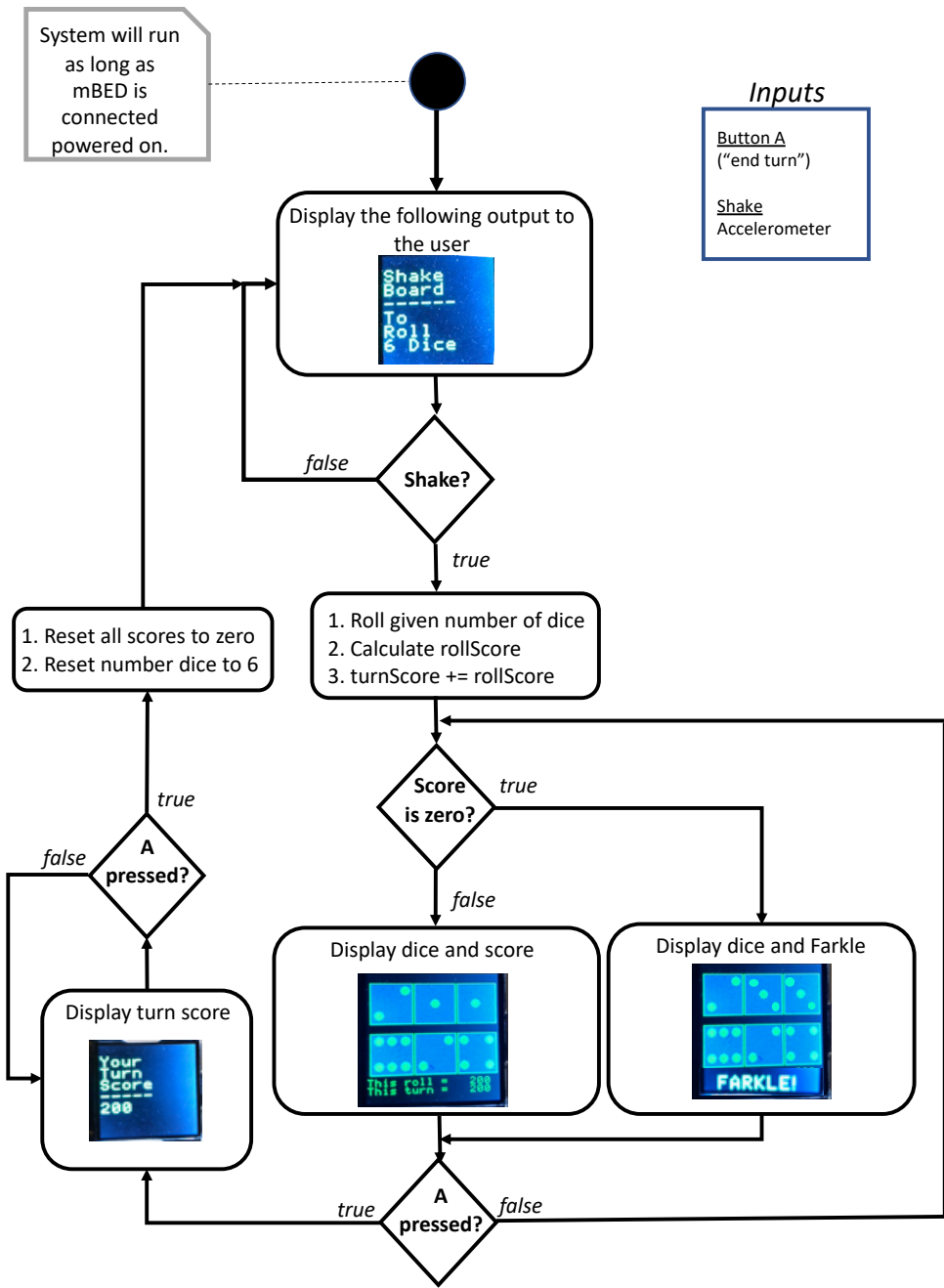
Stage 6: Create a hardware/software system that implements the specifications outlined in the activity (i.e. flowchart) chart in Appendix C. (+10 points added to your total)

Please make a YouTube video documenting this phase to receive a maximum final grade of 100 points, and upload the link to Canvas with a label of Stage6. Please note that if you complete Stage 6 then you do not have to show a video for Stage 1 thru Stage 5. Make sure you show yourself playing your game for at least 20 turns in a variety of different results. Make sure that we see a Farkle condition if possible.

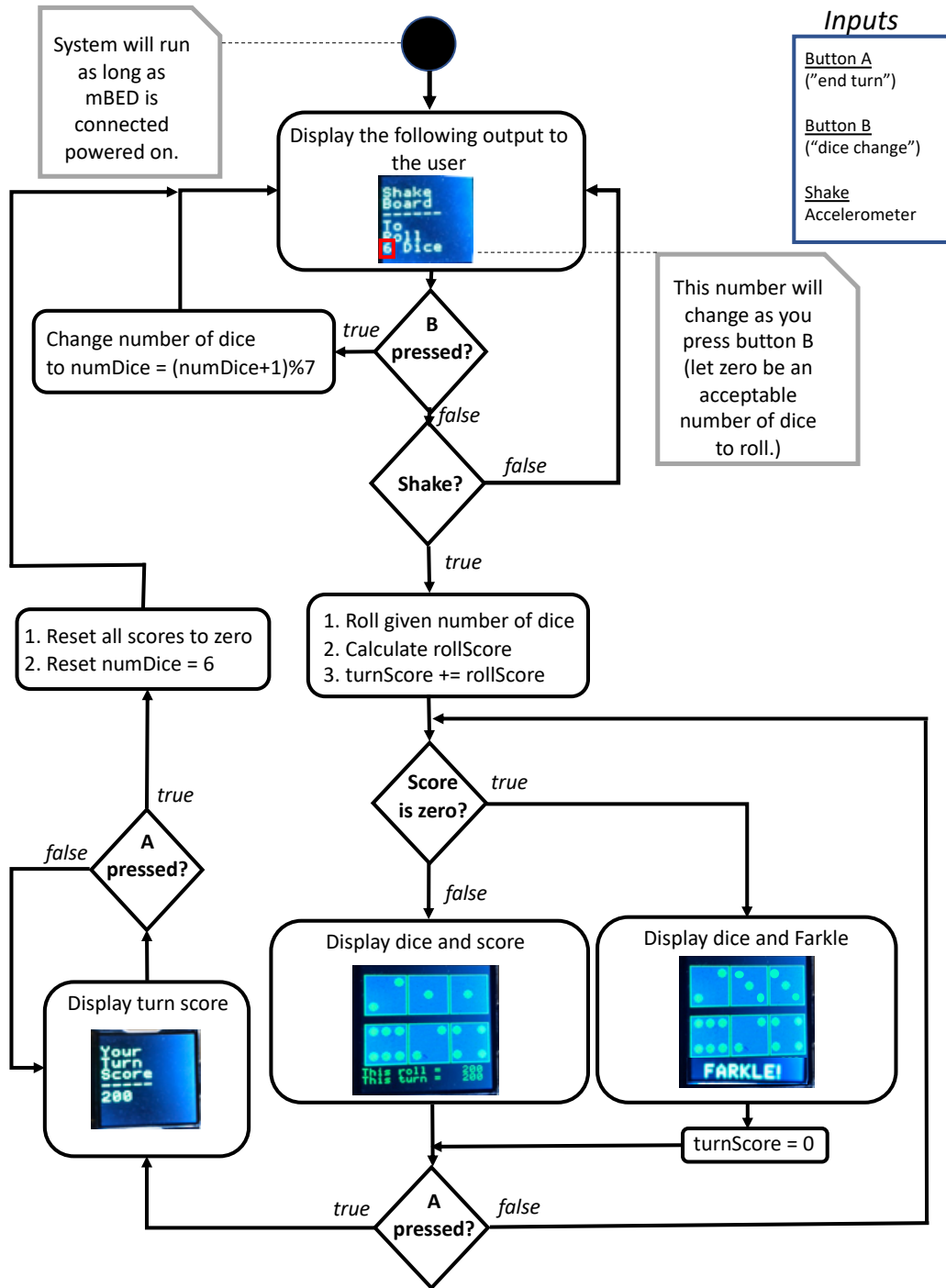
Remember the Farkle Assistant only keeps track of the current player's score, it does not have to keep track of the entire game in this version. You can use your project to enter a class Farkle competition that I will talk about at a later date.

Code Submission: *In addition to the video links, please make sure you upload your source code (i.e. header files, class source files, and main) to Canvas. You can upload a zipped file if you would like.*

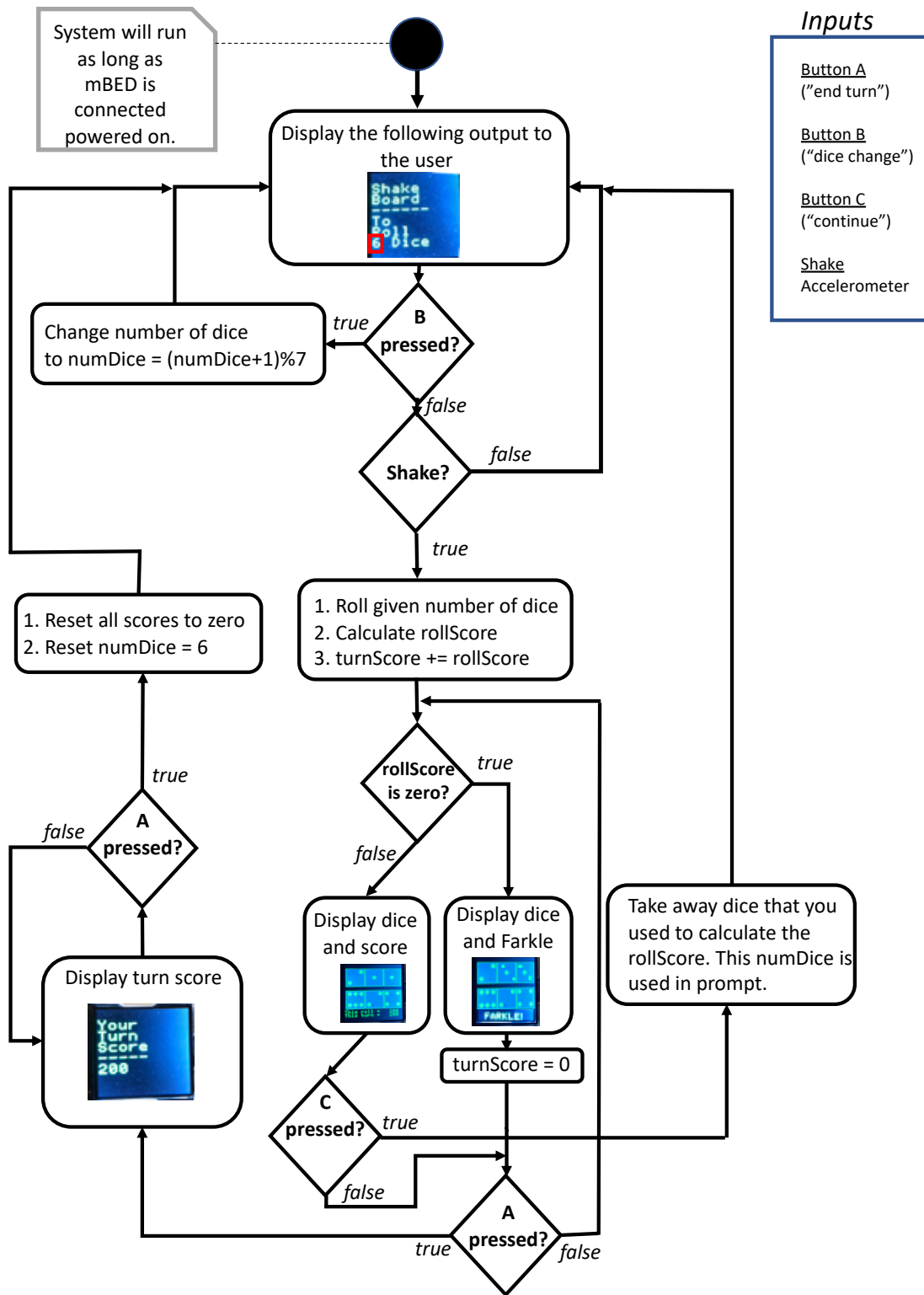
Appendix A – (max score is 80)



Appendix B – (max score is 90)



Appendix C – (max score is 100)



APPENDIX D

Grade Sheet Checkoff

Name of Student: _____ **Name of TA:** _____

(You may check off with the TA directly over video chat OR you may submit YouTube videos according to the specifications in the Lab Description.)

Completed Phase 1 Requirements (20%): _____

Completed Phase 2 Requirements (20%): _____

Completed Phase 3 Requirements (20%): _____

Completed Phase 4 Requirements (20%): _____

Completed Phase 5 Requirements (10%): _____

Completed Phase 6 Requirements (10%): _____

Additional points can be *deducted* at the discretion of the TA according to the following criteria.

Element	Percentage Deduction	Details
Had less than 2 classes in the implementation	15%	This is a requirement for the program. You can use more classes of course!
Did not use self-documenting coding styles	5%-15%	This can include incorrect indentation, using unclear variable names, or unclear comments.(See next page)
Did not upload source code to Canvas	10%	You can have the file zipped if you would like.

TURN IN EARLY POLICY

Element	Percentage Extra Credit	Details
Turn in on or before Oct 14	4%	Your final submission date off with TA or video submission
Turn In Oct 15	3%	Your final submission date off with TA or video submission
Turn In Oct 16	2%	Your final submission date off with TA or video submission
Turn In Oct 17	1%	Your final submission date off with TA or video submission

LATE POLICY

Element	Percentage Deduction	Details
Each Additional Day	15% per day	The weekend (Sat/Sun) will count as one day

APPENDIX E

Mbed Graphics and Timer Library and Constants.

This is available online through mbed.org, but I thought I would include a few items that might be useful to your system design.

These are constants that define the colors that you can use in this project.

```
#define WHITE 0xFFFFFFFF
#define BLACK 0x000000
#define RED   0xFF0000
#define GREEN 0x00FF00
#define BLUE  0x0000FF
#define LGREY 0xBF5F5F
#define DGREY 0x5F5F5F
```

These are some of the member functions that you can use with your uLCD_4DGL objects that you instantiate. The pixel range on the μ LCD-144-G2 is 128 pixels by 128 pixels.

```
void uLCD_4DGL :: pixel(int x, int y, int color)  // draw a pixel
void uLCD_4DGL :: line(int x1, int y1 , int x2, int y2, int color)
void uLCD_4DGL :: filled_circle(int x, int y , int radius, int color)
void uLCD_4DGL :: circle(int x, int y , int radius, int color)  // draw a circle in (x,y)
void uLCD_4DGL :: rectangle(int x1, int y1 , int x2, int y2, int color)  // draw a rectangle
void uLCD_4DGL :: filled_rectangle(int x1, int y1 , int x2, int y2, int color)
```

These are some of the timer functions that are available with the mBED system that are:

```
void  start ()
      Start the timer.
void  stop  ()
      Stop the timer.
void  reset ()
      Reset the timer to 0.
float read ()
      Get the time passed in seconds.
int   read_ms ()
      Get the time passed in mili-seconds.
int   read_us ()
      Get the time passed in micro-seconds.
```


Printing larger font example code can be accomplished with the following code. You should be able to find documentation online for each of these functions.

```
uLCD.cls();
uLCD.display_control(PORTRAIT);
uLCD.text_width(2);
uLCD.text_height(2);
uLCD.locate(0,1);
uLCD.printf("Shake\n");
uLCD.printf("Board\n");
uLCD.printf("-----\n");
uLCD.printf("To\n");
uLCD.printf("Roll\n");
```

Appendix F : Farkle Rules and Scoring.

(Please note that I have simplified the operation in the flowcharts in Appendix A-C, however, I would like for you to be able to calculate the correct score.)



Object

The player with the highest score above 10,000 points on the final round of play wins!

How to Play

Each player takes turns rolling the dice. When it's your turn, you roll all six dice at the same time. Points are earned every time you roll a 1 or 5, three of a kind, three pairs, a six-dice straight (1,2,3,4,5,6), or two triplets.

If none of your dice earned points, that's a Farkle! Since you earned no points, you pass the dice to the next player.

If you rolled at least one scoring die, you can bank your points and pass the dice to the next player, or risk the points you just earned during this round by putting ~~some or~~ all of the winning die (dice) aside and rolling the remaining dice. The remaining dice may earn you additional points, but if you Farkle, you lose everything you earned during the round.

Scoring is based only on the dice in each roll. You cannot earn points by combining dice from different rolls. You can continue rolling the dice until you either Pass or Farkle. Then the next player rolls the six dice until they Pass or Farkle. Play continues until it is your turn again.

The final round starts as soon as any player reaches 10,000 or more points.

Scoring Dice (in a single roll):

	= 50 pts		= 1500 pts
	= 100 pts		
	= 1000 pts	3 Pairs = 1500 pts	
	= 200 pts	2 Triplets = 2500 pts	
	= 300 pts	4 of a Kind = 1000 pts	
	= 400 pts	5 of a Kind = 2000 pts	
	= 500 pts	6 of a Kind = 3000 pts	
	= 600 pts	3 Farkles in a Row = Lose 1000 pts	